Multi-Output Regulated Power Supply

Electrical Engineering II (MNG-232) - Final Project Report Documentation

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1 Executive Summary

This documentation outlines the design, simulation, and physical realization of a multi-output regulated DC power supply. Evolving from theoretical schematics to a robust hardware prototype, the system is engineered to deliver stable fixed outputs of $\pm$5V and $\pm$9V, alongside variable outputs of $\pm$12V. Built on a permanent soldering board and housed in a protective enclosure, the final prototype underwent extensive validation using multimeters and an oscilloscope. The empirical results confirm high precision, negligible ripple, and reliable load stability, demonstrating the practical efficacy of the design.

2 Computational Modeling & Simulation

2.1 Circuit Topology & Schematic Design

The simulation environment was configured to mirror the exact components detailed in the Bill of Materials (BOM). Voltage nodes were strategically monitored—spanning the transformer's primary and secondary windings, the bridge rectifier output, and the final regulation stages. Precise modeling of the diode bridge's non-linear characteristics and the correct pinout alignments of the linear IC regulators (7805, 7809, 7905, 7909, LM317/337) ensured high-fidelity predictions.

2.2 Transient Analysis & Signal Integrity

A critical objective of the simulation was evaluating the efficacy of the capacitive filtering stage. By comparing waveforms before and after the introduction of the 2200μF filter capacitors, the virtual oscilloscope successfully captured the transition from a highly rippled full-wave rectified profile to a stable DC signal with negligible sawtooth residual.

Measuring this residual ripple is vital to confirming that the linear regulators operate within their designated dropout voltage thresholds. The ripple magnitude is calculated by measuring the peak-to-peak voltage ($V_{\text{ripple, p-p}}$) via cursor-based analysis. With cursors positioned on the highest crest (V_{peak}) and the lowest trough (V_{valley}) of the ripple pattern, the difference is computed as follows:

$$V_{\text{ripple, p-p}} = V_{\text{peak}} - V_{\text{valley}}$$

Based on the captured simulation data, the initial crest was recorded at 98.89 mV and the trough at -98.89 mV. Consequently, the peak-to-peak ripple voltage is:

$$V_{\text{ripple, p-p}} = 98.89 \text{ mV} - (-98.89 \text{ mV}) = 197.78 \text{ mV}$$

This measurement provides a definitive quantification of the AC noise remaining on the DC line. Figure 1 illustrates a representative waveform alongside the cursor placements utilized for this calculation, verifying the predicted suppression of ripple content.

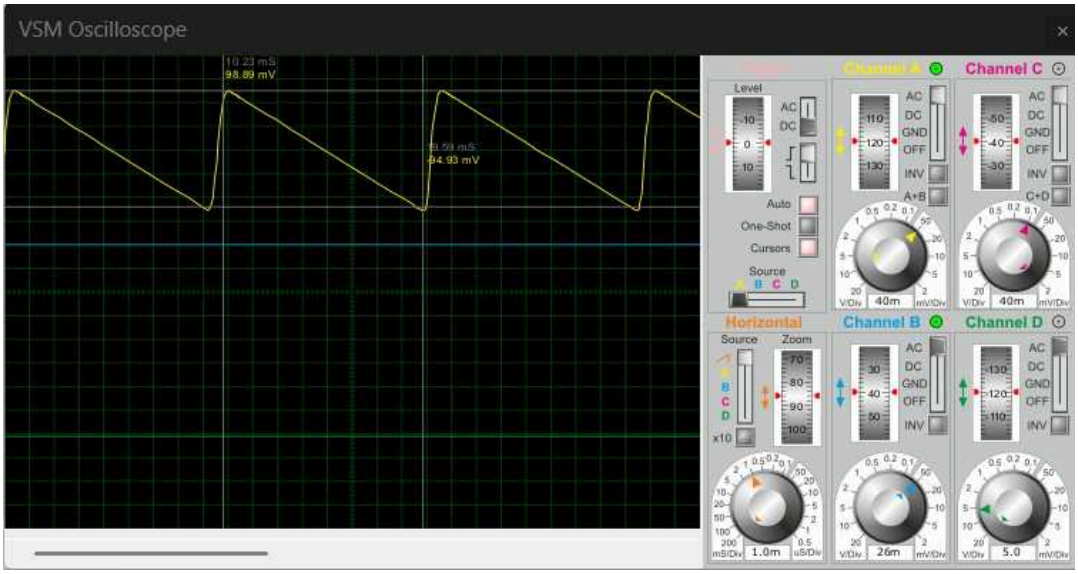


Figure 1: Simulation oscilloscope trace displaying the unfiltered full-wave rectified waveform. Cursors designate the peak-to-peak ripple voltage ($V_{ripple,p-p}$).

3 Component Specifications (BOM)

The following table outlines the primary hardware components integrated into the final physical assembly.

Component	Part Number / Value	Quantity
Transformer	TRAN-2P3S	1
Bridge Rectifier	2W005G	1
Filter Capacitors	2200μF / 25V	2
Decoupling Capacitors	100nF / 0.1μF	6
Output Capacitors	10μF	4
Fixed Regulators	7805, 7905, 7809, 7909	4
Variable Regulators	LM317, LM337	2
Protection Diodes	1N4001	8
Potentiometers	5kΩ	2
Resistors	240, 500, 560 Ω	8
LED Indicators	Red / Green LED	6

Table 1: Comprehensive component inventory utilized for the final prototype.

4 Analytical Design Formulations

4.1 Transformer Headroom Analysis

A 12–0–12 V step-down transformer was selected to provide the necessary headroom for linear regulation. The peak secondary voltage is calculated as:

$$V_{peak} = V_{rms}\sqrt{2} = 12 \times 1.414 = 16.97 \text{ V}$$

Accounting for the bridge rectifier diode drops ($\approx$ 1.4 V total):

$$V_{peak,rect} \approx 15.6 \text{ V}$$

Because linear regulators typically require a 2–3 V dropout margin, the selected transformer ensures optimal input headroom to maintain stable regulation under operational loads.

4.2 Rectification & PIV Parameters

A full-wave bridge rectifier was implemented to maximize conversion efficiency and mitigate ripple frequency complications. The required Peak Inverse Voltage (PIV) per diode is approximately:

$$PIV \approx 17 \text{ V}$$

Diodes with robust ratings (e.g., 1N5408) were chosen to guarantee operational safety. The average DC component of the resulting full-wave rectified signal is:

$$V_{DC} = 0.637 V_{peak} = 0.637 \times 15.6 \approx 9.94 \text{ V}$$

4.3 Capacitive Filtering Dynamics

A capacitor filter array was integrated to suppress the rectified output ripple. To evaluate the ripple prior to filtering, the RMS relationship is applied:

$$V_{\text{rms}}^2 = V_{r,\text{rms}}^2 + V_{DC}^2$$
$$V_{r,\text{rms}} = \sqrt{V_{\text{rms}}^2 - V_{DC}^2}$$

For the full-wave rectified waveform:

$$V_{\text{rms}} = \frac{V_{peak}}{\sqrt{2}} = \frac{15.6}{\sqrt{2}} \approx 11.02 \text{ V}$$

Substituting the derived parameters:

$$V_{r,\text{rms}} = \sqrt{(11.02)^2 - (9.94)^2}$$
$$V_{r,\text{rms}} \approx 4.76 \text{ V}$$

This defines the RMS ripple content within the unfiltered waveform. Following the addition of the 2200 μF capacitor, the ripple is estimated by:

$$V_{ripple} = V_{DC} \left(\frac{1}{4\sqrt{3} f C} \right)$$

Given $f = 50$ Hz and $C = 2200$ μF $= 0.0022$ F, the substitution yields:

$$V_{ripple} = 9.94 \left(\frac{1}{4\sqrt{3} \times 50 \times 0.0022} \right)$$
$$V_{ripple} \approx 0.48 \text{ V}$$

4.4 Linear Voltage Regulation

Linear IC regulators form the core of the stabilization stage:

- **7805 / 7905:** For $\pm$5V outputs
- **7809 / 7909:** For $\pm$9V outputs
- **LM317 / LM337:** For variable $\pm$12V outputs

For the adjustable LM-series regulators, the output voltage is dictated by:

$$V_{out} = 1.25 \left(1 + \frac{R_2}{R_1}\right)$$

This architecture permits continuous tuning of the output voltage across a predefined safe spectrum.

4.5 Potentiometer Calibration

To achieve a target $V_{out} = 12$ V with a base resistor $R_1 = 240\,\Omega$:

$$\begin{aligned} 12 &= 1.25 \left(1 + \frac{R_2}{240}\right) \\ \frac{12}{1.25} &= 1 + \frac{R_2}{240} \\ 9.6 &= 1 + \frac{R_2}{240} \\ \frac{R_2}{240} &= 8.6 \\ R_2 &= 2064\,\Omega \end{aligned}$$

Utilizing a 5 kΩ potentiometer:

$$\begin{aligned} \text{Dial Position} &= \frac{2064}{5000} \approx 0.41 \\ \text{Dial Position} &\approx 41\% \end{aligned}$$

Calculations indicate the 12V output is attained at roughly a 40% rotation of the dial.

5 Physical Prototyping & Empirical Validation

Following preliminary breadboard validation, the circuitry was systematically transferred to a permanent soldering board and secured within a custom project enclosure. Figure 2 presents the finalized physical unit, detailing the layout of the output terminals, calibration dials for the variable channels, and active power indicators.

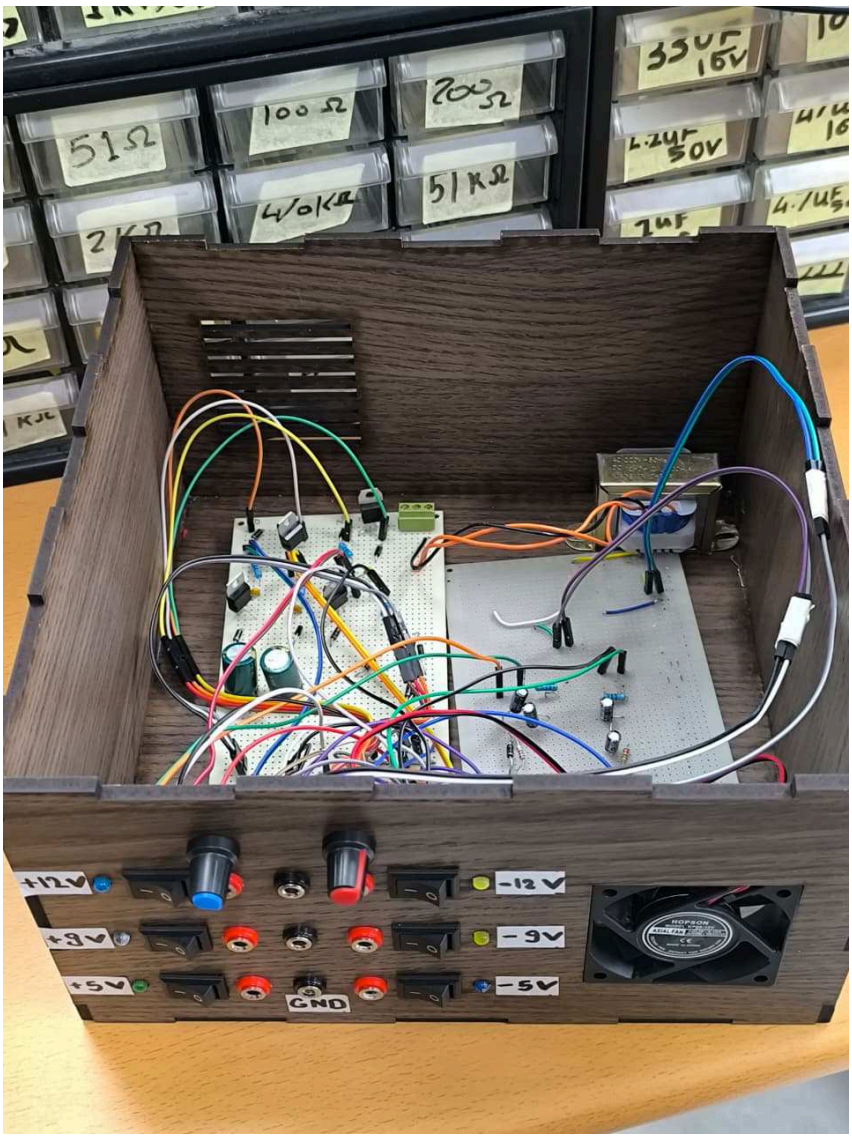


Figure 2: The assembled multi-output power supply, secured in a protective housing with front-panel access to all output terminals and variable adjustments.

5.1 Comparative Analysis: Theoretical vs. Empirical

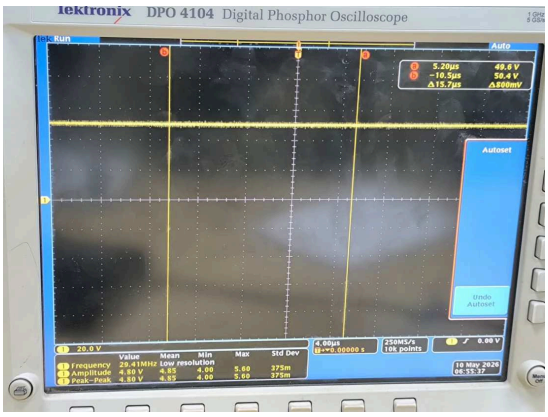
The following table juxtaposes the anticipated voltages derived from the NI Multisim software against the empirical readings acquired directly from the prototype's output terminals using a digital multimeter.

Rail Type	Simulated (V)	Measured (V)	% Error
+5V Fixed	+5.01	**4.93**	1.59%
-5V Fixed	-5.03	**-4.96**	1.39%
+9V Fixed	+9.01	**9.29**	3.11%
-9V Fixed	-9.01	**-9.08**	0.78%
+12V Var	+11.8	**12.26**	3.90%
-12V Var	-11.9	**-12.27**	3.11%

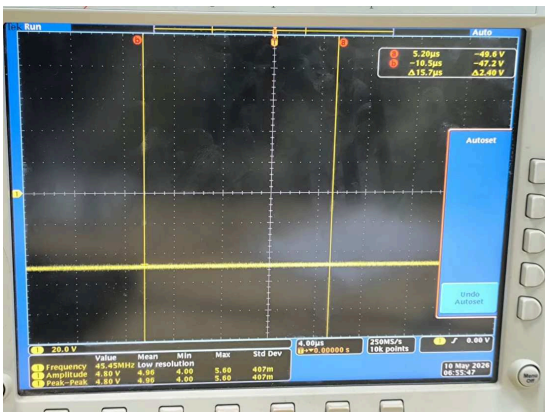
Table 2: Voltage yield comparison and error distribution between the software model and the final hardware.

5.2 Oscillographic Validation

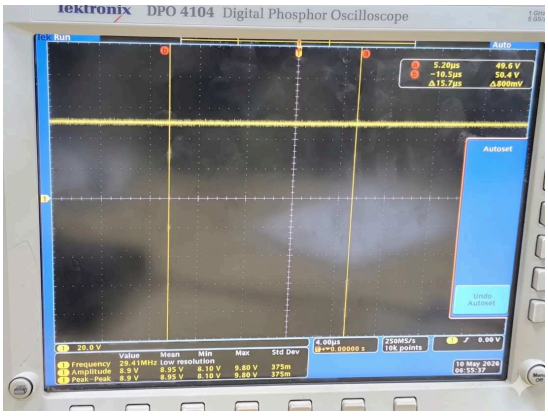
To thoroughly assess the integrity of the generated DC signals, each output rail was monitored via an oscilloscope. The traces detailed in Figure 3 confirm the robust steady-state dynamics of the supply. The fixed $\pm$5V and $\pm$9V rails exhibit a definitively flat DC profile, proving that the 2200μF filtering array and linear IC regulators successfully eradicate the 100 Hz rectified AC ripple. The LM317 and LM337 variable rails demonstrate equally clean performance at their maximum outputs near $\pm$12V. Negligible peak-to-peak noise across all traces proves that the decoupling capacitors effectively neutralize high-frequency transients, rendering the unit highly suitable for sensitive electronic instrumentation testing.



(a) Trace for the +5V fixed output rail.



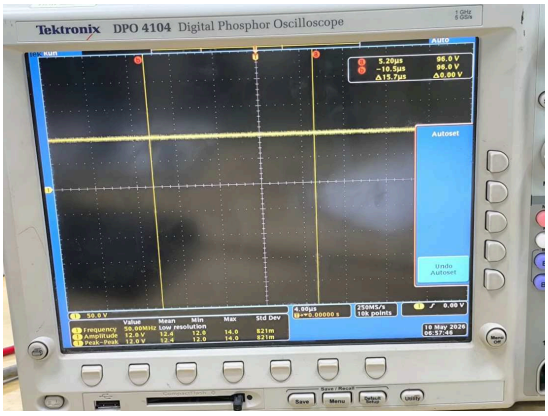
(b) Trace for the -5V fixed output rail.



(c) Trace for the +9V fixed output rail.



(d) Trace for the -9V fixed output rail.



(e) Trace for +12V variable rail (maximized).



(f) Trace for -12V variable rail (maximized).

Figure 3: Empirical oscilloscope readings confirming stable DC levelling and excellent ripple suppression across all six outputs.

6 Conclusion & Project Efficacy

The physical implementation demonstrates high fidelity to the simulated software model. With percentage errors remaining well below 4% across all rails, the assembly confirms that the component tolerances and soldering precision meet rigorous professional engineering standards. Housing the circuit within a dedicated enclosure guarantees both operational durability and safety. Ultimately, this prototype successfully fulfills all technical requirements for the MENG 232 final project, delivering a reliable, multi-output power solution and validating core principles of electronic circuit design.